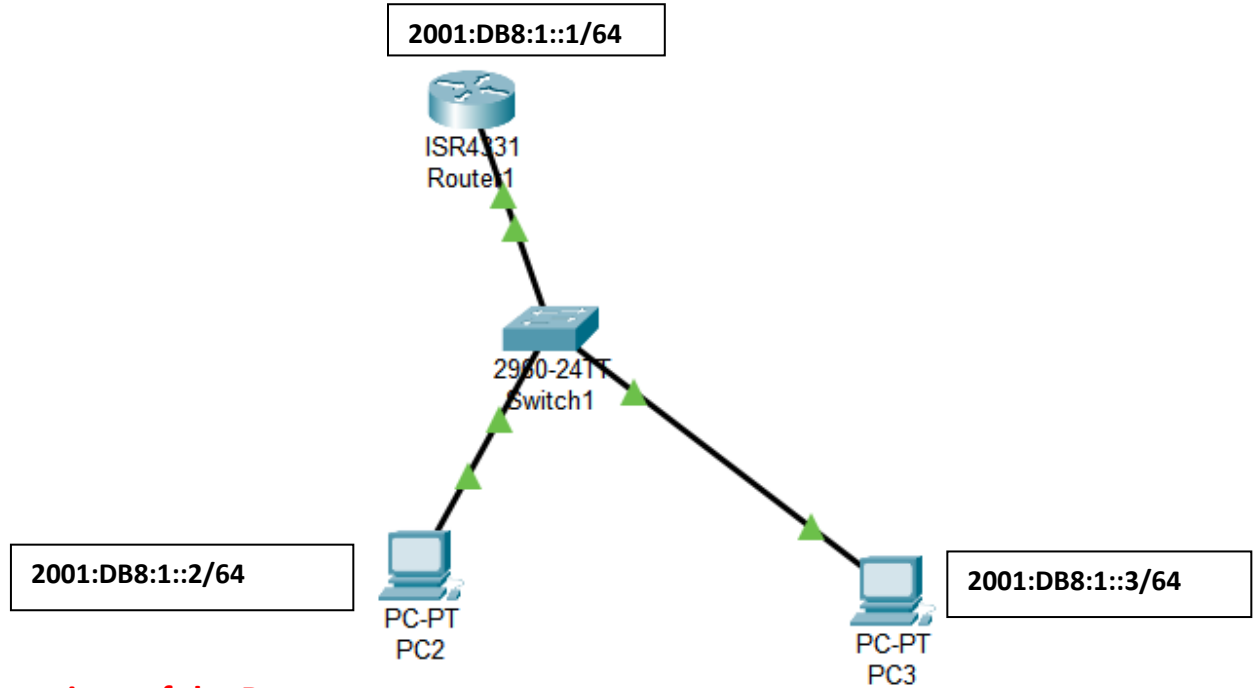


Lab 1:- Configurations of Static Routing Addressing in IPv6 Network



Configurations of the Router

- ✓ enable
- ✓ configure terminal
- ✓ interface GigabitEthernet0/0/0
- ✓ ipv6 address 2001:DB8:1::1/64
- ✓ no shutdown
- ✓ ipv6 unicast-routing
- ✓ exit
- ✓ write memory

Verification step

On PC2, go to **Command Prompt** and type:

ping 2001:DB8:1::3/64

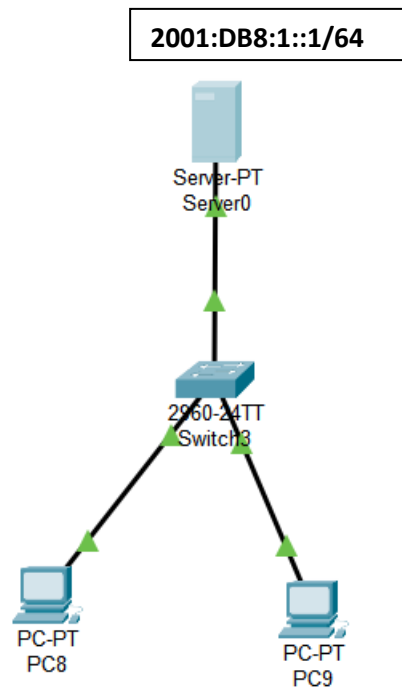
On PC3, try:

ping 2001:DB8:1::2/64

If configured correctly, you'll see successful ICMPv6 replies

Command	Meaning
enable	Enters privileged EXEC mode (admin access).
configure terminal	Enters global configuration mode to make system-wide changes.
interface GigabitEthernet0/0/0	Selects the specific interface to configure (in this case, GigabitEthernet0/0/0).
ipv6 address 2001:DB8:1::1/64	Assigns the IPv6 address 2001:DB8:1::1 with a subnet prefix length of /64 to the interface.
no shutdown	Activates the interface (by default, interfaces are shut down).
ipv6 unicast-routing	Enables IPv6 routing on the router so it can forward IPv6 packets.
exit	Exits the current configuration mode (interface or global).
write memory	Saves the running configuration to the startup configuration (so it's retained after reboot).

LAB 2 :- DHCPv6 Server Configuration



Configure the DHCPv6 Server

1. Click on the **DHCPv6 Server**.
2. Go to the **Services** tab.
3. Select **DHCPv6** from the left panel.
4. Configure the following settings:
 - o **Service:** Click **On**.
 - o **IPv6 Address Pool Name:** IPv6-Pool.
 - o **Prefix:** 2001:DB8:1::/64.
 - o **DNS Server:** 2001:4860:4860::8888 (Google's public IPv6 DNS).
 - o **Domain Name:** example.com.
5. Click **Save**.

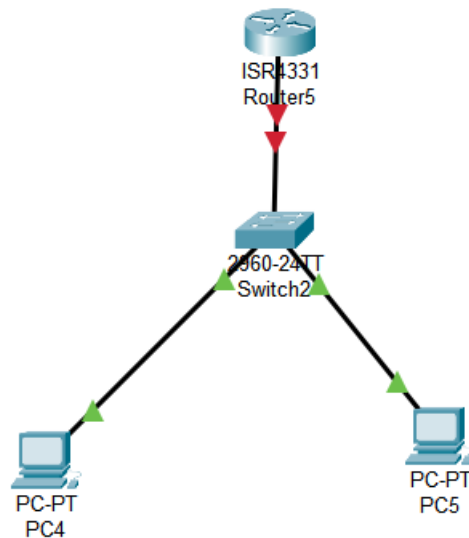
Configure Clients (PC1 and PC2) for DHCPv6

1. Click on **PC1**.
2. Go to **Desktop** → **IP Configuration**.
3. Select **Auto Config (IPv6)**.
4. Repeat the same steps for **PC2**.

LAB 3:- IPV6 SLAAC Configuration

Addressing Scheme (Example):

- **Router5 to Switch2 Link:** 2001:DB8:1:1::/64
- **PC4 and PC5 Network (SLAAC):** 2001:DB8:2:10::/64 (We'll use this prefix for SLAAC)



Configuration Steps:

1. Router5 Configuration

```
Router> enable
Router# configure terminal
Router(config)# ipv6 unicast-routing // Enable IPv6 routing
Router(config)# interface GigabitEthernet0/0 // Interface connected to Switch2
Router(config-if)# ipv6 address 2001:DB8:1:1::1/64 // Assign IPv6 address to interface
Router(config-if)# ipv6 enable // Enable IPv6 on the interface (required for SLAAC)
Router(config-if)# ipv6 nd other-config-flag // Optional: Set the "Other Config" flag
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# end
Router# copy running-config startup-config // Save configuration
```

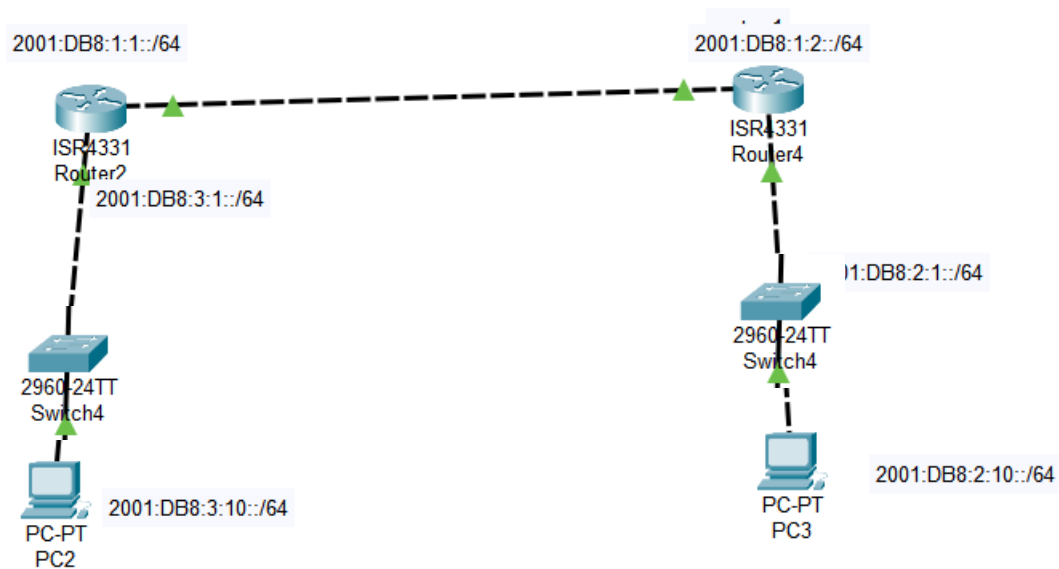
PC4 and PC5 Configuration (SLAAC):

- **IPv6 Configuration:** Set to "Auto-Config" or "Stateless DHCP" (depending on Packet Tracer version). This tells the PCs to use SLAAC to obtain an IPv6 address.

Verification:

- **PC IPv6 Address:** Check the IPv6 address on PC4 and PC5 using `ipconfig /all` (Windows) or `ifconfig` (Linux/macOS) to verify they received an address via SLAAC. You should see an address starting with 2001:DB8:2:10::.

LAB 4:- Configuration of RIPng in IPv6



Configuration Steps:

1.Router2 Configuration:

```
Router> enable
Router# configure terminal
Router(config)# ipv6 unicast-routing // Enable IPv6 routing
Router(config)# interface GigabitEthernet0/0 // Interface to Router1
Router(config-if)# ipv6 address 2001:DB8:1:1::1/64
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface GigabitEthernet0/1 // Interface to Switch0
Router(config-if)# ipv6 address 2001:DB8:2:1::1/64
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ipv6 router rip RIPNG // Enable RIPng
Router(config-rtr)# redistribute connected // Redistribute connected routes
Router(config-rtr)# exit
```

2.Router4 Configuration:

```
Router> enable
Router# configure terminal
Router(config)# ipv6 unicast-routing
Router(config)# interface GigabitEthernet0/0 // Interface to Router0
Router(config-if)# ipv6 address 2001:DB8:1:1::2/64
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface GigabitEthernet0/1 // Interface to Switch1
Router(config-if)# ipv6 address 2001:DB8:3:1::1/64
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ipv6 router rip RIPNG
Router(config-rtr)# redistribute connected
Router(config-rtr)# exit
Router(config)# end
```

Enable RIPng in Both Routers(Router 2 and Router 4

```
Router(config)#interface GigabitEthernet0/0/0
```

```
Router(config-if)#ipv6 rip RIPNG enable
```

```
Router(config-if)#exit
```

```
Router(config)#interface GigabitEthernet0/0/1
```

```
Router(config-if)#ipv6 rip RIPNG enable
```

```
Router(config-if)#exit
```

PC2 Configuration (IPv6 Address and Gateway):

- **IPv6 Address:** 2001:DB8:2:10::10/64
- **IPv6 Gateway:** 2001:DB8:2:1::1

IT Can be static or may be Dynamic ip addressing

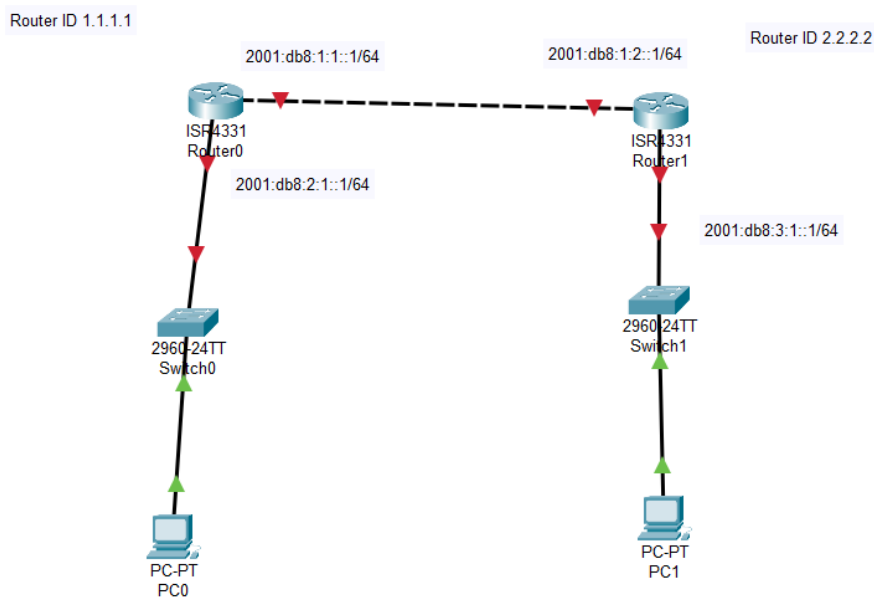
6. PC4 Configuration (IPv6 Address and Gateway):

- **IPv6 Address:** 2001:DB8:3:10::10/64
- **IPv6 Gateway:** 2001:DB8:3:1::1

Verification:

- **In Both Routers :** use *show ipv6 protocols*
- **Routing Table:** Use *show ipv6 route* on both routers to check the learned routes.
- **Connectivity:** Ping between PC2 and PC4 to verify end-to-end connectivity.

LAB 5:- Configuration of OSPFv3 in IPv6



For Router 1

Router>en

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ipv6 unicast-routing

Router(config)#**interface gigabitethernet0/0/0**

Router(config-if)#**ipv6 address 2001:db8:1:1::1/64**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

interface gigabitethernet0/0/1

Router(config-if)#**ipv6 address 2001:db8:1:2::1/64**

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

interface gigabitethernet0/0/0

Router(config-if)#**ipv6 ospf 1 area 0**

%OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id, please configure manually

Router(config-if)#**interface gigabitethernet0/0/1**

Router(config-if)#**ipv6 ospf 1 area 0**

Router(config-if)#**ipv6 router ospf 1**

Router(config-rtr)#**router-id 1.1.1.1**

Router(config-if)#**end**

write memory

For router 2

```
Router>en
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface gigabitethernet0/0/0
Router(config-if)#ipv6 address 2001:db8:1:2::1/64
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```

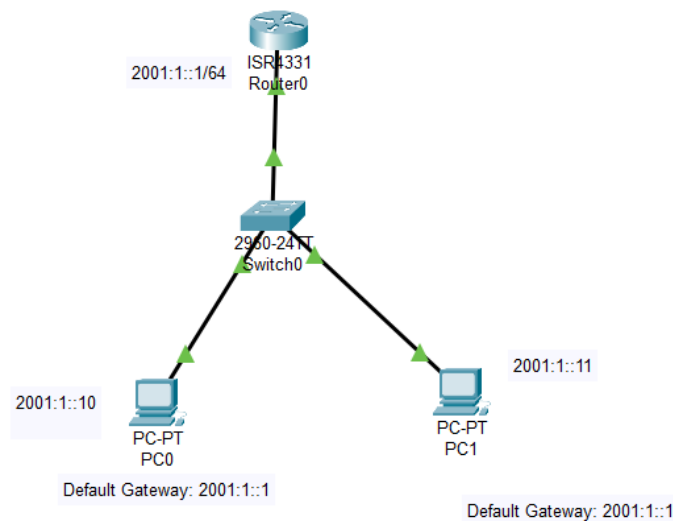
```
interface gigabitethernet0/0/1
Router(config-if)#ipv6 address 2001:db8:3:1::1/64
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
interface gigabitethernet0/0/0
Router(config-if)#ipv6 ospf 1 area 0
%OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id,please configure manually
Router(config-if)#interface gigabitethernet0/0/1
Router(config-if)#ipv6 ospf 1 area 0
Router(config-if)#ipv6 router ospf 1
Router(config-rtr)#router-id 2.2.2.2
Router(config-if)#end
write memory
```

Verification Commands:

- ✓ **show ipv6 route**
- ✓ **show ipv6 route ospf**: Displays the IPv6 routes learned via OSPFv3.
- ✓ **show ipv6 ospf**: Shows general information about the OSPFv3 process.
- ✓ **show ipv6 ospf interface**

LAB 6:- ICMPv6 Configuration in Packet Tracer



ICMPv6 Configuration in Packet Tracer

Step 1: Enable IPv6 Routing on the Router

1. Click on **Router0**.
2. Go to the **CLI** tab.
3. Enter the following commands:

Router> enable

Router# configure terminal

Router(config)# ipv6 unicast-routing

Step 2: Configure IPv6 Addresses On Router0

Router(config)# interface gigabitEthernet 0/0

Router(config-if)# ipv6 address 2001:1::1/64

Router(config-if)# no shutdown

Router(config-if)# exit

1. On PC0

1. Click **PC0** > Desktop > **IPv6 Configuration**.
2. Uncheck **Auto Configuration**.
3. Set:
 - IPv6 Address: 2001:1::10
 - Subnet Prefix Length: 64
 - Default Gateway: 2001:1::1

2. On PC1

1. Click **PC1** > Desktop > **IPv6 Configuration**.
2. Uncheck **Auto Configuration**.
3. Set:

- IPv6 Address: 2001:1::11
- Subnet Prefix Length: 64
- Default Gateway: 2001:1::1

Verify ICMPv6 Using Ping

On **PC0**, go to **Command Prompt** and type:

```
ping 2001:1::11
```

On **PC1**, try:

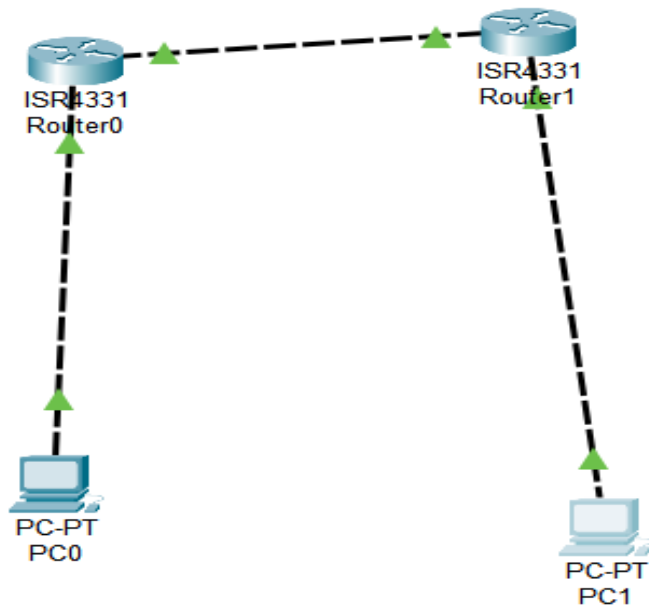
```
ping 2001:1::10
```

If configured correctly, you'll see successful ICMPv6 replies.

To see the ICMPv6 packets in more detail:

1. Click the **"Simulation Mode"** button (looks like a fast-forward icon with a pause button below the topology).
2. **Initiate a ping from PC0 to PC1** as described in Step 3.
3. **Observe the Events:** In the "Simulation Panel" at the bottom, you will see ICMPv6 events.
4. **Click on an ICMPv6 event** to view the Packet Details at different OSI layers.

LAB 7:- IPV6 to ipv4 tunneling



1.IP Addressing Plan

Device	Interface	IPv4	IPv6
PC0	NIC	192.168.10.2/24	2001:db8:10::2/64
Router0	G0/0	192.168.10.1/24	2001:db8:10::1/64
Router0	G0/1	10.0.0.1/30	2001:db8:1::1/64
Router1	G0/1	10.0.0.2/30	2001:db8:1::2/64
Router1	G0/0	192.168.20.1/24	2001:db8:20::1/64
PC1	NIC	192.168.20.2/24	2001:db8:20::2/64

2. Configure PC0 and PC1

Go to each PC > Desktop > IP Configuration

For PC0:

- IPv4 Address: 192.168.10.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- IPv6 Address: 2001:db8:10::2/64
- Default Gateway: 2001:db8:10::1

For PC1:

- IPv4 Address: 192.168.20.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1
- IPv6 Address: 2001:db8:20::2/64
- Default Gateway: 2001:db8:20::1

3. Configure Router0

```
enable
conf t

interface g0/0
ip address 192.168.10.1 255.255.255.0
ipv6 address 2001:db8:10::1/64
no shutdown

interface g0/1
ip address 10.0.0.1 255.255.255.252
ipv6 address 2001:db8:1::1/64
no shutdown

exit
ipv6 unicast-routing
```

4. Configure Router1

```
enable
conf t

interface g0/0
ip address 192.168.20.1 255.255.255.0
ipv6 address 2001:db8:20::1/64
no shutdown

interface g0/1
ip address 10.0.0.2 255.255.255.252
ipv6 address 2001:db8:1::2/64
no shutdown

exit
ipv6 unicast-routing
```

Verification Step

- From PC0:
 - Ping 192.168.20.2 (PC1)
 - Ping 2001:db8:20::2
- From PC1:
 - Ping 192.168.10.2 (PC0)
 - Ping 2001:db8:10::2

Use **ping** and **ipconfig** from PC's Command Prompt

LAB 8:- Capturing and Analyzing IPv6 Packets Using Wireshark

LAB 9:- IPv6 or IPv4 Header Examination with Wireshark”

Analysis

1. **Version**
2. **Traffic Class**
3. **Flow Label**
4. **Payload Length**
5. **Next Header**
6. **Hop Limit**
7. **Source IPv6/IPv4 Address**
8. **Destination IPv6/IPv4 Address**